

Kathryn Cantrell

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EDUCATION

Loyola University Chicago, MS *Data Science*

Expected Graduation: December 2026

Loyola University Chicago, BS *Applied Mathematics*

WORK EXPERIENCE

Loyola University Chicago Stritch School of Medicine

May 2025- Present

Research Assistant - Ji Lab

- Use real-world medical data to build predictive models relating to alcohol use, ARDS, and sarcoidosis
- Model-building using a wide variety of R and Python packages
- Results utilized in several papers under peer review

Tennessee DECA

May 2024- Present

Alumnus Student Consultant

- Provide data-driven consulting services to clients across multiple domains, including environmental science, industry startups, and financial modeling
- Collaborate with interdisciplinary teams to analyze complex datasets and develop actionable insights
- Design and implement predictive models using R, Python, and SQL; create custom visualizations to support decision-making
- Results used for secondary and post-secondary competition scenarios

National Science Foundation Undergraduate Researcher

May 2023- July 2024

Using Agent-Based Modeling to Understand Biofilm Growth and Eradication by Antibiotics and Phages Respectively

- Worked alongside a peer to develop a mathematical model to simulate a piece of the human small intestine to identify optimal treatment strategies
- Utilized NetLogo to create simulations for immunocompromised in vivo settings, immunocompetent in vitro settings, and in vitro settings
- Produced a manuscript to document the extensive background, the methodologies utilized, the results discovered, and a discussion surrounding the project
- Presented at the American Women in Mathematics Symposium in November 2023, Joint Mathematics Meeting in January 2024

LUROP Research Fellow

May 2022- May 2023

Poster Research Covering Evolutionary Game Theory

- Python-coded a random 10x10 grid filled with cells that depicted cooperative and noncooperative behaviors
 - Presented at the Mathematical Association of America's MAAthFest in August 2022 and National Conference of Undergraduate Research in April 2023
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Technical Skills

- Programming Languages: R, Python, SQL, NetLogo
- Data Analysis & Modeling: tidyverse, pandas, NumPy, scikit-learn, PyTorch, TensorFlow, ggplot2, matplotlib, seaborn, Plotly, Canva

- Machine Learning & Simulation: regression modeling, agent-based modeling, clustering, classification, differential equation modeling
- Productivity & Collaboration: Microsoft Office Suite (Excel, Word, PowerPoint, SharePoint), Google Workspace (Docs, Sheets, Slides, Drive)
- Tools & Platforms: Git, Jupyter, RStudio, Tableau, VS Code, SQLite, PostgreSQL (basic familiarity)
- Environments: Windows, Linux (basic command line)